

# An Introduction to Applied Bayesian Statistics with Stan

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2026-08-17



# Overview

- **Goals**

Provide the foundation and orientation so that you can

- Use Stan (or **brms**) in your work
- Understand and enjoy StanCon talks!

- **Outline**

- About Stan ✓
- Bayesian models and workflow ✓
- Model specification in Stan ✓
- Multilevel model specification in **brms** ✓
- HMC and multi-level models ←

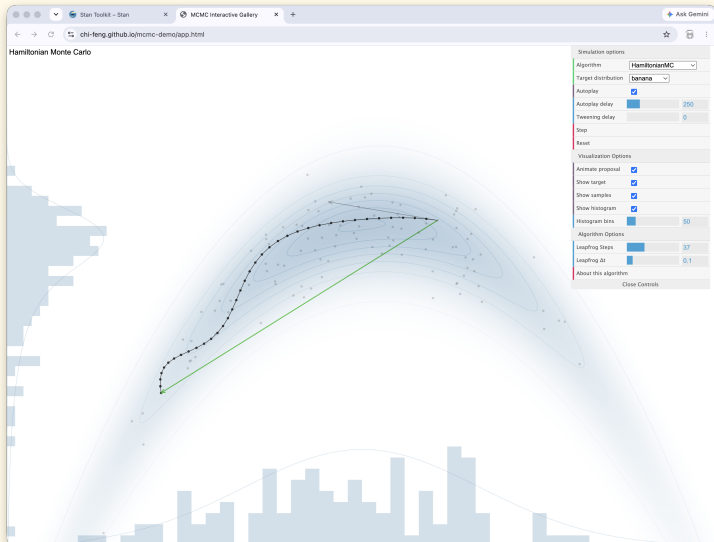
- **Slides** [stancon\\_2026\\_stan\\_intro](#)

# Hamiltonian Monte Carlo

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# Hamiltonian Monte-Carlo

MCMC Interactive Gallery - <https://chi-feng.github.io/mcmc-demo/app.html>



# Hamiltonian Monte-Carlo

- Generate draws from the **joint** log posterior density  $\log p(\theta, y)$ 
  - follow the **Hamiltonian flow** according to the **gradient** of the density
  - Stan's automatic-differentiation computes these gradients
- HMC computes on the unconstrained scale  $\mathbb{R}^N$ 
  - Stan provides constraining/unconstraining transforms
- The Hamiltonian flow is continuous; discretized to **steps**
  - One draw from the posterior consists of many steps
- HMC parameters must be tuned for the posterior geometry
  - Tune step size and estimate mass matrix during warmup, determine number of steps on-the-fly (NUTS)
- See Michael Betancourt's excellent [A Conceptual Introduction to Hamiltonian Monte Carlo](#)

# Euclidean Hamiltonian Monte Carlo

- **Phase space:**  $q$  position (parameters);  $p$  momentum
- **Target density:**  $\pi(q)$
- **Potential energy:**  $V(q) = -\log \pi(q)$
- **Metric (mass matrix):**  $M$ 
  - rescales and rotates the dynamics to match posterior geometry
- **Momentum refresh:**  $p \sim \text{MultiNormal}(0, M)$
- **Kinetic energy:**  $T(p) = \frac{1}{2} p^\top M^{-1} p$
- **Hamiltonian:**  $H(p, q) = V(q) + T(p)$

$$\frac{dq}{dt} = M^{-1} p$$

$$\frac{dp}{dt} = \nabla_q \log \pi(q)$$

# Leapfrog integrator

- Solves Hamilton's equations by **simulating dynamics**  
(symplectic [volume preserving];  $\epsilon^3$  error per step,  $\epsilon^2$  total error)
- Given: **step size**  $\epsilon$ , **mass matrix**  $M$ , **parameters**  $q$
- **Initialize kinetic** energy,  $p \sim \text{Normal}(0, \mathbf{I})$
- **Repeat** for  $L$  leapfrog steps:

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad [\text{half step in momentum}]$$

$$q \leftarrow q + \epsilon M^{-1} p \quad [\text{full step in position}]$$

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad [\text{half step in momentum}]$$

# HMC Tuning Parameters

- HMC simulate Hamiltonian trajectory of a particle through parameter space
  - at each iteration, provide a refresh of random momentum  $\rho$  where

$$\rho \sim \text{MultiNormal}(0, M),$$

where  $M$  is the positive definite mass matrix

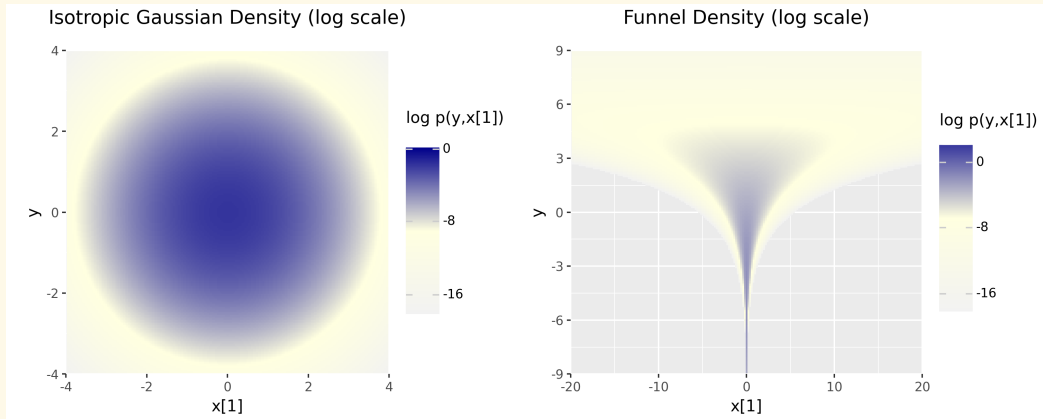
- this puts posterior directions on comparable scales  
(we have a high-dimensional posterior)
- The well-tuned HMC sampler requires that:
  - one trajectory is composed of just enough steps
  - the step-size just small enough to follow the Hamiltonian
  - the mass matrix is a good *linear pre-conditioner*



# Efficiency Tuning

- The HMC sampler explores the posterior density of the *parameter space*
- If the posterior is (approximately) multivariate normal the mass matrix produces an isotropic Gaussian posterior - a single step in every direction is accurate
- If the posterior is sharply curved, no single mass matrix is valid everywhere
  - during warmup the step-size is adjusted smaller and smaller
  - the sampler fails to make progress
- We can rewrite Stan programs in terms of *transformed parameters*
  - see the Stan User's Guide [Efficiency Tuning](#) chapter.
- We have already seen this: the centered and non-centered parameterizations of a hierarchical model

# Efficiency Tuning for Hierarchical Models



*Easy vs. hard posterior densities*

# Efficiency Tuning for Hierarchical Models

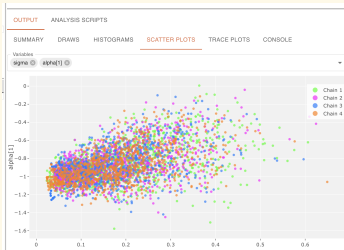
- The isotropic Gaussian density is *not* a hierarchical model
  - it is the product of  $N$  normal distributions
- The funnel distribution is hierarchical
  - the scale of elements  $x$  depends on global parameter  $y$
- **But** *the posterior density is the product of the prior times the likelihood*
  - shape of the posterior morphs as the model sees more data
- You see this already in the Stan Playground scatter plots

# Baseball Example Posterior Densities

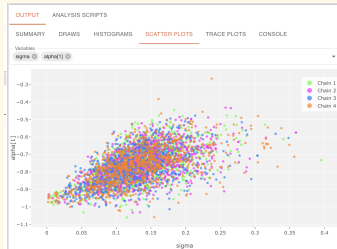
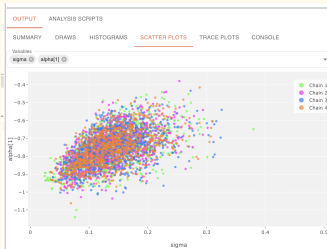
Centered parameterization

Non-centered parameterization

Low

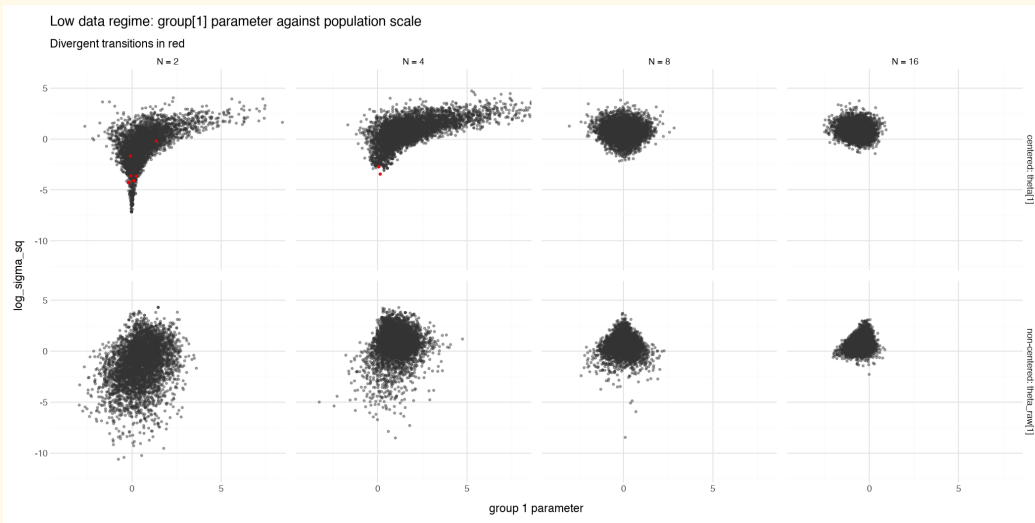


Higher

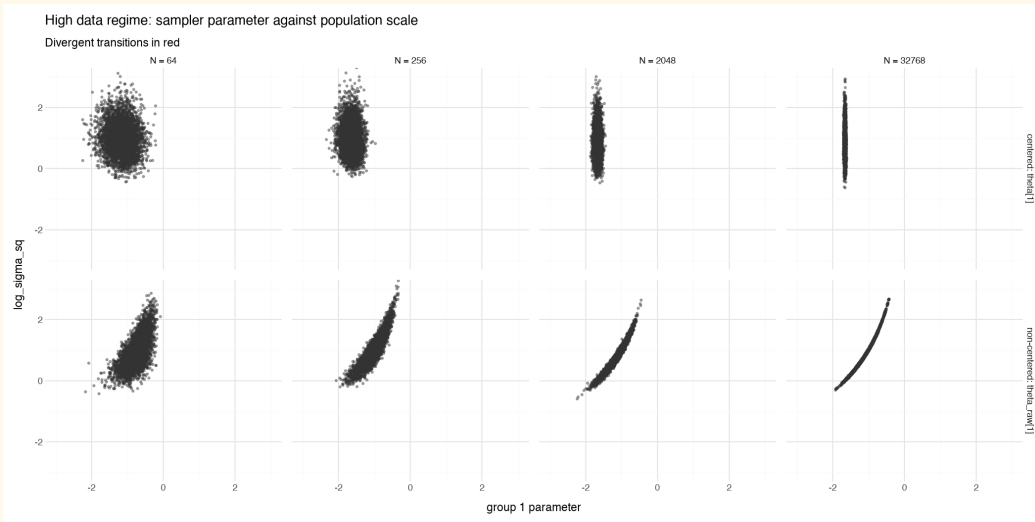


- The R-hat statistic and ESS are related
  - see [Nested R-hat: Assessing the convergence of Markov chain Monte Carlo when running many short chains](#)
  - a further consequence is slow sampling time (low ESS/s) as well.
- In low data regimes the non-centered parameterization improves performance
  - better ESS
  - faster total sampling time
- In high data regimes the centered parameterization wins
- The baseball data is an illustration of “low data” and “more data”

# Centered vs. Non-Centered Parameterization: Small Data



# Centered vs. Non-Centered Parameterization: Big Data



# Centered vs. Non-Centered Parameterization: Complicated Data

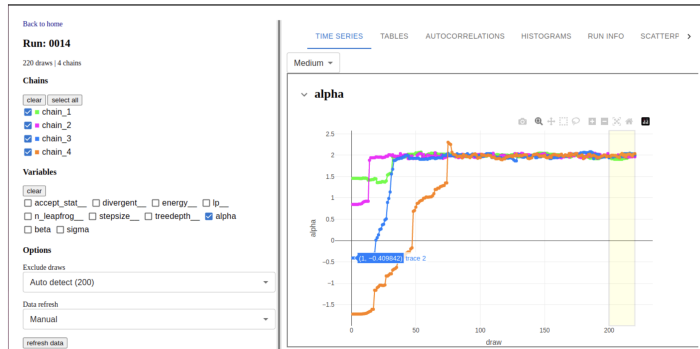
- These plots were generated from synthetic data generated according to the model
- Beyond a few simple cases, there is no one good solution
- *Note:* **brms** always uses the non-centered parameterization
- Some References
  - [A General Framework for the Parametrization of Hierarchical Models](#)  
Papaspiliopoulos et al, 2007
  - [Hamiltonian Monte Carlo for Hierarchical Models](#)  
Betancourt and Girolami, 2013
  - The Stan User's Guide chapters:  
[Problematic Posteriors](#), [Reparameterizations](#) [Efficiency Tuning](#)



# Diagnosing Complications: the MCMC monitor

MCMC Monitor : <https://github.com/flatironinstitute/mcmc-monitor>

MCMC Monitor enables tracking and visualization of MCMC processes executed with [Stan](#) in local or remote web browsers. When you run a sampler, you can configure Stan to generate output to a directory on your computer. MCMC Monitor reads this output and displays it in the web app, with real-time updates. As you track the progress of the run, MCMC provides diagnostic plots and statistics.



## Questions and Discussion

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# Stan Documentation

- Stan Documentation Suite
  - [Stan User's Guide](#)
  - [Reference Manual](#) - just the facts
  - [Functions Reference](#) - Stan math library functions available in the Stan language
- Learning Resources
  - [Tutorials](#) - just a few of many
  - [Stan Case Studies](#) - best practices, deep dives for many models
  - [StanCon Talks](#) - past StanCons
  - [Field Guides](#) - education, ecology, epidemiology, cogsci, pharma
- [Stan Forums](#) - your questions answered